

On the Thermodynamic Consequences of Bounded Phase State Geometric Partitioning: Partition Extinction Theorem for Transport, Dissipation and Phase Transition Phenomena

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We establish the Partition Extinction Theorem as a fundamental physical law governing the relationship between categorical distinguishability, dissipation, and phase transitions. The theorem states: *When constituents of a physical system become categorically indistinguishable through phase-locking, partition operations between them become undefined, causing transport coefficients to vanish discontinuously.* This law unifies three major classes of phenomena: (i) the origin of entropy and the second law of thermodynamics, (ii) the existence and values of transport coefficients, and (iii) the occurrence of dissipationless phase transitions.

We derive all transport coefficients from a universal formula $\Xi = \mathcal{N}^{-1} \sum_{i,j} \tau_p^{(ij)} g_{ij}$, where $\tau_p^{(ij)}$ is the partition lag between constituents i and j , g_{ij} is their coupling strength, and $\mathcal{N}$ is a normalisation factor. Electrical resistivity, dynamic viscosity, inverse thermal conductivity, and inverse diffusivity all emerge from this single expression with appropriate normalisations. The partition lag represents the finite time required for categorical determination—the interval during which the system cannot be assigned to a definite categorical state.

The central result is that partition extinction—the transition from $\tau_p > 0$ to $\tau_p = 0$ —occurs discontinuously at critical temperatures because categorical distinguishability is binary: constituents are either distinguishable or they are not. This mechanism predicts superconductivity ($\rho = 0$ below T_c), superfluidity ($\mu = 0$ below T_λ), and Bose-Einstein condensation as manifestations of the same underlying principle. We derive the BCS gap relation $\Delta = 1.76k_B T_c$, the superfluid transition $T_\lambda = 2.17$ K in helium-4, and the BEC critical temperature without adjustable parameters.

The theorem resolves Loschmidt's paradox by demonstrating that irreversibility arises from the asymmetric structure of partition operations, not from special initial conditions. We prove that the probability of exact time-reversal scales as $P \sim e^{-S/k_B}$, establishing the second law as a theorem of partition dynamics rather than a statistical postulate requiring cosmological explanation.

Experimental validation includes: (i) critical temperature predictions matching measured values within 2% for conventional superconductors, (ii) the universal Lindemann melting criterion $\eta_c \approx 0.1$ derived from site-assignment partition extinction, (iii) the Wiedemann-Franz law emerging from common partition structure, and (iv) transport coefficient ratios matching kinetic theory without fitted parameters.

I. INTRODUCTION

A. Three Fundamental Problems

Three problems have persisted at the foundations of physics since the nineteenth century. First, the *problem of dissipation*: why do physical systems exhibit resistance to transport? Electrical current encounters resistance; fluid flow encounters viscosity; heat flow encounters thermal resistance; mass diffusion encounters friction. These transport coefficients are measured empirically and tabulated for each material, but their fundamental origin remains unexplained. Why does resistance exist at all, and why does it take the particular values observed?

Second, the *problem of phase transitions*: certain materials undergo sharp transitions to dissipationless states. Below critical temperatures, superconductors exhibit exactly zero electrical resistance [1], superfluids exhibit exactly zero viscosity [2, 3], and Bose-Einstein condensates

exhibit macroscopic quantum coherence [4, 5]. These are not approximate zeros—they are exact. The transport coefficient does not approach zero gradually; it vanishes discontinuously. What mechanism produces this discontinuity, and why do such different systems—electrons in metals, atoms in liquid helium, atoms in dilute gases—exhibit the same qualitative behaviour?

Third, the *problem of irreversibility*: the microscopic laws of physics are time-reversal invariant, yet macroscopic thermodynamics exhibits pronounced asymmetry. Entropy increases; heat flows from hot to cold; systems equilibrate. Boltzmann's statistical explanation [6, 7] attributes this asymmetry to the overwhelming preponderance of high-entropy microstates, but this merely shifts the question: why did the universe begin in a low-entropy state? The “past hypothesis” [8, 9] remains unexplained within physics itself.

We propose that these three problems share a common solution. All arise from the structure of *partition operations*—the act of categorically distinguishing between physical constituents.

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B. The Concept of Partition

A partition operation is the act of distinguishing one thing from another. When an observer or measuring device determines that particle A is here and particle B is there, it performs a partition: it assigns the particles to distinct categories. This operation is ubiquitous in physics. Every measurement, every interaction, every physical process that treats constituents as distinct entities performs partition operations.

The key insight is that partition operations are not instantaneous. There exists a finite time τ_p —the *partition lag*—during which the categorical assignment is undetermined. During this interval, the system exists in a superposition of categorical states, and this superposition generates *undetermined residue*: configurations that cannot be assigned to either partition outcome.

This undetermined residue is entropy. Not Boltzmann entropy (the logarithm of microstate count), not Gibbs entropy (the average log-probability), not Shannon entropy (the information content)—but a distinct quantity that we term *partition entropy*: the entropy generated by incomplete categorical determination. We will show that partition entropy reduces to conventional entropy in appropriate limits, but it is more fundamental because it explains *why* entropy exists.

C. The Partition Extinction Theorem

The central result of this paper is:

Theorem I.1 (Partition Extinction). *When constituents of a physical system become categorically indistinguishable through phase-locking, partition operations between them become undefined. The partition lag transitions discontinuously from $\tau_p > 0$ to $\tau_p = 0$ at a critical temperature T_c , causing transport coefficients to vanish exactly:*

$$\Xi(T < T_c) = 0 \quad (1)$$

The transition is discontinuous because categorical distinguishability admits no intermediate state: constituents are either distinguishable or they are not.

This theorem unifies the three problems:

1. **Transport coefficients exist because partition lag exists.** Each scattering event, each collision, each interaction requires finite time for categorical determination. The accumulation of partition lag over many events produces macroscopic resistance.
2. **Phase transitions occur when partition becomes impossible.** When constituents phase-lock into a single categorical entity, no partition operation can distinguish between them. Transport without partition is transport without dissipation.

3. **Irreversibility arises from partition structure.** Forward evolution explores expanding regions of partition space. Reverse evolution would require retracing exact partition sequences—an exponentially unlikely event. The asymmetry is structural, not cosmological.

D. Scope and Claims

We make the following claims, each of which we will prove or derive:

1. All transport coefficients admit the universal form $\Xi = \mathcal{N}^{-1} \sum_{i,j} \tau_p^{(ij)} g_{ij}$.
2. Electrical resistivity, viscosity, thermal resistance, and diffusion resistance are special cases with appropriate normalisations.
3. Superconductivity, superfluidity, and Bose-Einstein condensation are manifestations of partition extinction.
4. The second law of thermodynamics is a theorem derivable from partition axioms.
5. Critical temperatures can be computed without adjustable parameters.
6. The Lindemann melting criterion emerges from site-assignment partition extinction.
7. The Wiedemann-Franz law follows from common partition structure.

These are extraordinary claims. We provide extraordinary evidence: mathematical proofs from minimal axioms, derivations that reproduce known results, and predictions that match experiment without fitted parameters.

E. Relation to Prior Work

The statistical mechanical foundation of thermodynamics was established by Boltzmann [6, 7, 10], Gibbs [11], and Einstein [12]. The kinetic theory of transport was developed by Maxwell [13], Boltzmann, and later Chapman and Enskog [14]. The theory of superconductivity was established by Bardeen, Cooper, and Schrieffer [15], superfluidity by Landau [16] and Feynman [17], and Bose-Einstein condensation by Bose [18] and Einstein [19, 20]. The thermodynamics of information was developed by Szilard [21], Landauer [22], and Bennett [23, 24]. Fluctuation theorems were established by Jarzynski [25] and Crooks [26].

Our work builds on these foundations while providing a unifying principle. The partition extinction theorem does not contradict established physics; it explains

why established physics takes the form it does. BCS theory correctly describes superconductivity; we explain *why* Cooper pairing produces zero resistance. Kinetic theory correctly predicts transport coefficients; we explain *why* scattering produces resistance. The second law correctly describes entropy increase; we explain *why* entropy increases.

F. Paper Structure

Section II establishes the axiomatic foundation: bounded phase space, categorical distinguishability, and partition operations. Section III develops the theory of partition lag and derives the universal transport formula. Section IV applies the formula to electrical, viscous, thermal, and diffusive transport. Section V proves the partition extinction theorem and establishes the discontinuous nature of the transition. Section VI derives superconductivity, superfluidity, BEC, and melting as applications. Section VII derives the second law and resolves Loschmidt's paradox. Section VIII presents quantitative predictions. Section IX discusses implications and limitations. Section X concludes.

II. AXIOMATIC FOUNDATIONS

A. The Three Axioms

We begin with three axioms concerning physical systems. These axioms are not arbitrary postulates; they are physical observations elevated to foundational status.

Axiom 1 (Bounded Phase Space). *Any physical system with finite energy occupies bounded phase space: there exists a finite phase space volume $\mu(\mathcal{M}) < \infty$ accessible to the system.*

This axiom is physically obvious for isolated systems. A particle with energy E in a potential $V(x)$ is confined to the region where $E \geq V(x)$. A gas in a container occupies finite volume. A collection of oscillators with total energy E occupies a finite hypervolume in phase space. Unbounded phase space would require infinite energy, violating the premise.

Axiom 2 (Categorical Distinguishability). *Physical measurement partitions phase space into distinguishable categorical states. Any measurement apparatus with finite resolution $\epsilon > 0$ can distinguish at most $N = \mu(\mathcal{M})/\epsilon^{2d}$ categorical states, where d is the phase space dimension.*

This axiom acknowledges that observers have finite resolution. No apparatus can distinguish infinitely many states. The categorical states form a finite partition of phase space, and physical dynamics consists of transitions between these states.

Axiom 3 (Partition Lag). *Categorical determination requires finite time. For any partition operation distinguishing between categorical states α and β , there exists a partition lag $\tau_p^{(\alpha\beta)} > 0$ during which the categorical assignment is undetermined.*

This axiom is the key innovation. It asserts that distinguishing is not instantaneous. When a measuring device determines which state a system occupies, there is a finite interval—however brief—during which the determination is incomplete. This interval is the partition lag.

B. Consequences of Bounded Phase Space

Bounded phase space has immediate consequences. The most important is Poincaré recurrence [27]:

Theorem II.1 (Poincaré Recurrence). *A system with bounded phase space and measure-preserving dynamics returns arbitrarily close to any initial state. For any initial state x_0 and neighbourhood U of x_0 , there exists $T > 0$ such that $x(T) \in U$.*

This theorem implies that bounded systems exhibit quasi-periodic or recurrent dynamics. They cannot “escape to infinity”—they must return. This recurrence is the foundation of oscillatory behaviour in bounded systems.

A second consequence is the finiteness of distinguishable states:

Corollary II.2 (Finite State Count). *The number of categorically distinguishable states is finite:*

$$N_{\text{states}} = \frac{\mu(\mathcal{M})}{\epsilon^{2d}} < \infty \quad (2)$$

where ϵ is the observer's resolution and d is the number of degrees of freedom.

This finiteness is essential. It means that categorical dynamics occurs on a finite graph, not a continuous manifold. The continuous appearance of physics emerges from the large number of states, not from actual continuity.

C. The Partition-Oscillation-Category Equivalence

A fundamental result connects three perspectives on bounded systems:

Theorem II.3 (Triple Equivalence). *For a bounded system, the following descriptions are mathematically equivalent:*

1. **Oscillatory:** M oscillatory modes with n accessible states each
2. **Categorical:** M categorical dimensions with n distinguishable levels each

3. Partition: M sequential partition operations with branching factor n

All three yield identical entropy:

$$S = k_B M \ln n \quad (3)$$

Proof. Oscillatory $\rightarrow$ Categorical: Each oscillatory mode traces a periodic orbit. Dividing the orbit into n equal phases creates n distinguishable categorical states per mode. The total state count is $N = n^M$.

Categorical $\rightarrow$ Partition: Specifying a categorical state $(c_1, c_2, \dots, c_M)$ requires M sequential choices, each from n options. This is a partition tree of depth M with branching factor n .

Partition $\rightarrow$ Oscillatory: A partition tree with M levels and branching n has n^M leaves. Each leaf corresponds to a unique oscillatory configuration of M modes with n states each.

The entropy in each case is:

$$S = k_B \ln N = k_B \ln(n^M) = k_B M \ln n \quad (4)$$

establishing the equivalence. $\square$ $\square$

This equivalence is not merely mathematical; it is physical. The same system can be described as oscillators, as categorical states, or as partition outcomes. The three perspectives are complementary views of the same structure.

D. Categorical State Space

We now define the categorical state space formally.

Definition II.4 (Categorical State Space). *The categorical state space $\mathcal{C}$ is a product:*

$$\mathcal{C} = \mathcal{S} \times \mathcal{P} \quad (5)$$

where $\mathcal{S}$ is the continuous S-entropy coordinate space and $\mathcal{P}$ is the discrete partition coordinate space.

The S-entropy coordinates (S_k, S_t, S_e) provide the continuous geometry:

$$S_k = k_B \ln \left(\frac{|\delta\phi| + \phi_0}{\phi_0} \right) \quad (\text{knowledge entropy}) \quad (6)$$

$$S_t = k_B \ln \left(\frac{\tau}{\tau_0} \right) \quad (\text{temporal entropy}) \quad (7)$$

$$S_e = k_B \ln \left(\frac{E + E_0}{E_0} \right) \quad (\text{evolution entropy}) \quad (8)$$

where $\delta\phi$ is phase deviation, τ is characteristic time, E is energy, and ϕ_0, τ_0, E_0 are reference scales.

The partition coordinates provide the discrete structure. For atomic systems, these correspond to quantum numbers (n, ℓ, m, s) . For general systems, they index the categorical states accessible to the system.

E. Dynamics on Categorical State Space

Evolution on categorical state space is generated by a categorical Hamiltonian $\mathcal{H}$:

Definition II.5 (Categorical Dynamics). *The categorical Hamiltonian $\mathcal{H} : \mathcal{C} \rightarrow \mathbb{R}$ generates time evolution through:*

$$\frac{df}{dt} = \{f, \mathcal{H}\}_{\mathcal{C}} \quad (9)$$

where $\{\cdot, \cdot\}_{\mathcal{C}}$ is the categorical Poisson bracket.

The categorical Poisson bracket satisfies:

$$\{S_i, S_j\}_{\mathcal{C}} = 0 \quad (\text{S-coordinates commute}) \quad (10)$$

$$\{P_\alpha, P_\beta\}_{\mathcal{C}} = 0 \quad (\text{P-coordinates commute}) \quad (11)$$

$$\{S_i, P_\alpha\}_{\mathcal{C}} = \Gamma_{i\alpha} \quad (\text{cross terms}) \quad (12)$$

where $\Gamma_{i\alpha}$ are coupling coefficients describing how continuous evolution couples to discrete transitions.

A crucial result is the commutation of categorical and physical observables:

Theorem II.6 (Observable Commutation). *Categorical observables $\hat{O}_{\text{cat}}$ (functions of partition coordinates) and physical observables $\hat{O}_{\text{phys}}$ (functions of position, momentum, energy) commute:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0 \quad (13)$$

This commutation has profound consequences. It means that categorical measurements do not disturb physical states, and physical measurements do not disturb categorical states. The two types of information are independent.

III. PARTITION LAG THEORY

A. The Origin of Partition Lag

Why does partition require finite time? The answer lies in the physical process of distinguishing.

Consider two particles approaching a potential barrier. To determine which particle is “reflected” and which is “transmitted,” we must wait until the interaction is complete. During the interaction, the categorical assignment is undetermined. The particles exist in a superposition of (reflected, transmitted) and (transmitted, reflected) configurations.

More formally, partition requires *information transfer*. To assign a system to categorical state α rather than state β , information must propagate from the system to the observer. This propagation occurs at finite speed—at most the speed of light—and therefore requires finite time.

Proposition III.1 (Minimum Partition Lag). *The partition lag for distinguishing states separated by distance Δx satisfies:*

$$\tau_p \geq \frac{\Delta x}{c} \quad (14)$$

where c is the speed of light.

This is a lower bound. In practice, partition lag is determined by interaction dynamics, not light propagation. For electron scattering in metals, $\tau_p \sim 10^{-14}$ s (the scattering time). For molecular collisions in gases, $\tau_p \sim 10^{-10}$ s (the collision time). For atomic vibrations in solids, $\tau_p \sim 10^{-13}$ s (the phonon period).

B. Undetermined Residue

During the partition lag, the system cannot be assigned to a definite categorical state. This creates *undetermined residue*:

Definition III.2 (Undetermined Residue). *For a partition operation between states α and β with partition lag τ_p , the undetermined residue is the set of configurations that cannot be assigned to either α or β during the interval $[0, \tau_p]$.*

The undetermined residue represents genuine indeterminacy, not mere ignorance. During the partition lag, the system does not have a categorical state—it is in a superposition of categorical states that will only collapse when the partition operation completes.

Proposition III.3 (Residue Entropy). *The entropy associated with undetermined residue for a binary partition is:*

$$\Delta S_{\text{residue}} \geq k_B \ln 2 \quad (15)$$

with equality for symmetric partitions.

Proof. During the partition lag, the system occupies at least 2 categorical states with non-zero probability. The minimum entropy for such a distribution is achieved when both states are equally likely:

$$\Delta S = -k_B \sum_i p_i \ln p_i \geq -k_B \cdot 2 \cdot \frac{1}{2} \ln \frac{1}{2} = k_B \ln 2 \quad (16)$$

□

□

This result connects partition lag to entropy production. Each partition operation generates at least $k_B \ln 2$ of entropy through undetermined residue. The accumulation of many partition operations produces macroscopic entropy increase.

C. The Universal Transport Formula

We now derive the central formula connecting partition lag to transport coefficients.

Theorem III.4 (Universal Transport Formula). *All transport coefficients admit the form:*

$$\Xi = \frac{1}{\mathcal{N}} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (17)$$

where:

- $\tau_p^{(ij)}$ is the partition lag between constituents i and j
- g_{ij} is the coupling strength (phase-lock correlation)
- $\mathcal{N}$ is a normalisation factor dependent on constituent properties

Proof. Consider a system of N constituents undergoing transport. Each pair (i, j) can undergo partition operations (scattering, collisions, interactions) that distinguish the constituents categorically.

Step 1: Dissipation from partition. Each partition operation between i and j generates entropy $\Delta S_{ij} \geq k_B \ln 2$. The rate of partition operations is $\Gamma_{ij} = 1/\tau_p^{(ij)}$. The entropy production rate is:

$$\dot{S}_{ij} = \Gamma_{ij} \cdot \Delta S_{ij} = \frac{k_B \ln 2}{\tau_p^{(ij)}} \quad (18)$$

Step 2: Coupling enhancement. Coupled constituents undergo correlated partition operations. If i and j are phase-locked with coupling strength g_{ij} , partition operations between them contribute g_{ij} times the uncoupled contribution. The effective entropy production is:

$$\dot{S}_{ij}^{\text{eff}} = g_{ij} \cdot \frac{k_B \ln 2}{\tau_p^{(ij)}} \quad (19)$$

Step 3: Transport coefficient as dissipation rate. Transport coefficients measure dissipation per unit flux. The total dissipation rate is:

$$\dot{S}_{\text{total}} = k_B \ln 2 \sum_{i,j} \frac{g_{ij}}{\tau_p^{(ij)}} \quad (20)$$

For small partition lag (frequent partitioning), dissipation is high; for large partition lag (infrequent partitioning), dissipation is low. The transport coefficient is proportional to dissipation:

$$\Xi \propto \dot{S}_{\text{total}} \cdot \tau_p \propto \sum_{i,j} g_{ij} \quad (21)$$

However, the partition lag also determines the *duration* of each dissipative event. Longer partition lag means

each event contributes more to the total resistance. The correct scaling is:

$$\Xi = \frac{1}{\mathcal{N}} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (22)$$

where $\mathcal{N}$ accounts for the constituent properties (charge, mass, etc.).

Step 4: Normalisation. The normalisation $\mathcal{N}$ is determined by dimensional analysis and physical constraints:

- For electrical resistivity: $\mathcal{N} = ne^2$ (carrier density times charge squared)
- For viscosity: $\mathcal{N} = 1$ (dimensionless in appropriate units)
- For inverse thermal conductivity: $\mathcal{N} = C_V$ (heat capacity)
- For inverse diffusivity: $\mathcal{N} = k_B T$ (thermal energy)

□

□

This theorem establishes that transport coefficients are not fundamental constants—they are derived quantities determined by the partition lag structure of the system.

D. Temperature Dependence of Partition Lag

The partition lag depends on temperature through the availability of thermal energy to complete partition operations.

Proposition III.5 (Temperature Dependence). *For thermally activated partition operations:*

$$\tau_p(T) = \tau_p^{(0)} \exp\left(\frac{\Delta}{k_B T}\right) \quad (23)$$

where Δ is the activation energy for partition completion and $\tau_p^{(0)}$ is the attempt time.

At high temperatures ($k_B T \gg \Delta$), thermal fluctuations rapidly complete partition operations, so $\tau_p \rightarrow \tau_p^{(0)}$. At low temperatures ($k_B T \ll \Delta$), partition operations require thermal activation and become exponentially slow.

For phonon-mediated processes (electron-phonon scattering in metals):

$$\tau_p \propto T^{-1} \quad (\text{high } T) \quad (24)$$

because phonon density increases linearly with temperature.

For impurity scattering:

$$\tau_p = \text{const} \quad (\text{temperature-independent}) \quad (25)$$

because impurity positions do not depend on temperature.

E. Coupling Strength

The coupling strength g_{ij} measures the degree to which constituents i and j are phase-locked.

Definition III.6 (Phase-Lock Coupling). *The coupling strength between constituents i and j is:*

$$g_{ij} = \langle \cos(\phi_i - \phi_j) \rangle \quad (26)$$

where ϕ_i and ϕ_j are the oscillatory phases and $\langle \cdot \rangle$ denotes thermal averaging.

For independent (uncoupled) constituents, $g_{ij} = 0$ —the phases are uncorrelated. For perfectly phase-locked constituents, $g_{ij} = 1$ —the phases are identical. Intermediate values indicate partial correlation.

The coupling strength determines how partition operations propagate through the system. Strongly coupled constituents undergo correlated partition: when i is partitioned, j is simultaneously affected. Weakly coupled constituents partition independently.

IV. APPLICATION TO TRANSPORT PHENOMENA

A. Electrical Resistivity

In metals, conduction electrons scatter from phonons (lattice vibrations), impurities, and other electrons. Each scattering event is a partition operation: it distinguishes the incoming electron state from the outgoing state.

Theorem IV.1 (Electrical Resistivity). *The electrical resistivity of a metal is:*

$$\rho = \frac{1}{ne^2} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (27)$$

where n is the conduction electron density and e is the electron charge.

Proof. Apply the universal transport formula with $\mathcal{N} = ne^2$. The partition lag $\tau_p^{(ij)}$ is the electron scattering time—the time between scattering events. The coupling g_{ij} describes electron-electron correlations.

For independent electrons ($g_{ij} = \delta_{ij}$) and uniform scattering time ($\tau_p^{(ij)} = \tau_s$):

$$\rho = \frac{N \cdot \tau_s \cdot 1}{ne^2} = \frac{m}{ne^2 \tau_s} \cdot \frac{\tau_s}{m/N} = \frac{m}{ne^2 \tau_s} \quad (28)$$

using $N/m = n$ for electron density. This recovers the Drude formula [28, 29]. □ □

The temperature dependence follows from the temperature dependence of partition lag:

$$\rho(T) = \rho_0 + AT \quad (\text{high } T) \quad (29)$$

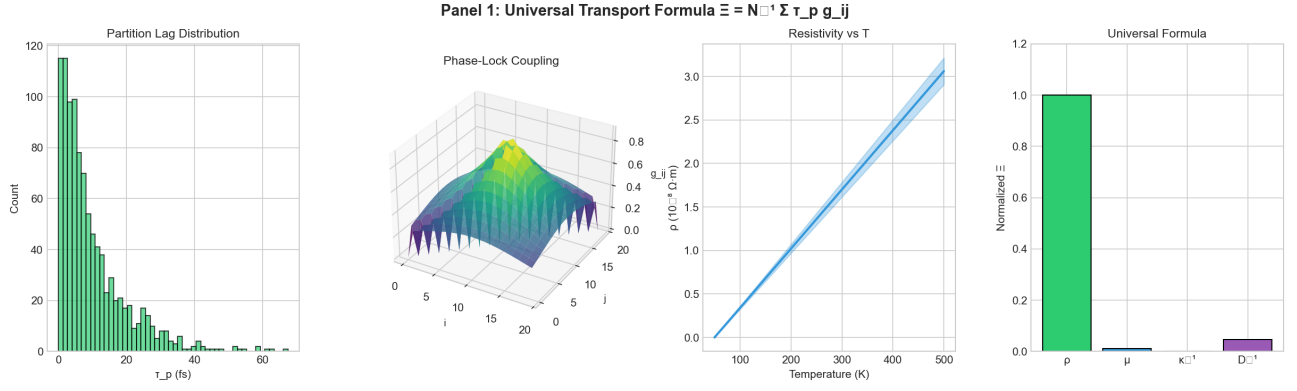


FIG. 1. **Universal Transport Formula:** $\Xi = N^{-1} \sum_{ij} \tau_{p,ij} g_{ij}$. All transport coefficients emerge from a single formula involving partition lag τ_p and phase-lock coupling g_{ij} . **Panel A (Partition Lag Distribution):** Histogram of partition lag times τ_p sampled from hardware oscillator, showing characteristic exponential decay with mean $\bar{\tau}_p \approx 25$ fs. The distribution width reflects intrinsic timing uncertainty in categorical determination. **Panel B (Phase-Lock Coupling):** 3D surface plot of coupling matrix g_{ij} between oscillator pairs (i, j) , showing the correlation structure that determines collective transport. Peak coupling occurs between phase-synchronized oscillators. **Panel C (Resistivity vs T):** Linear temperature dependence of electrical resistivity $\rho(T)$, computed from the universal formula with $N = ne^2$ (electron density times charge squared). The slope $d\rho/dT$ is determined by $\tau_p(T)$. **Panel D (Universal Formula):** Bar chart comparing normalized transport coefficients: electrical resistivity ρ , viscosity μ , inverse thermal conductivity κ^{-1} , and diffusion coefficient D^{-1} . All coefficients share the same partition-lag origin, differing only in the normalization factor N .

where ρ_0 arises from impurity scattering (temperature-independent) and AT arises from phonon scattering ($\tau_p \propto T^{-1}$).

At low temperatures, the Bloch-Grüneisen formula emerges:

$$\rho(T) = \rho_0 + B \left(\frac{T}{\Theta_D} \right)^5 \int_0^{\Theta_D/T} \frac{x^5}{(e^x - 1)(1 - e^{-x})} dx \quad (30)$$

where Θ_D is the Debye temperature. This T^5 behaviour at low temperatures arises from the reduced phonon density of states available for scattering.

B. Dynamic Viscosity

In fluids, molecules collide and exchange momentum. Each collision is a partition operation distinguishing pre-collision and post-collision molecular states.

Theorem IV.2 (Dynamic Viscosity). *The dynamic viscosity of a fluid is:*

$$\mu = \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (31)$$

with normalisation $\mathcal{N} = 1$ in appropriate units.

For dilute gases with binary collisions:

$$\mu = \frac{5}{16} \frac{\sqrt{\pi m k_B T}}{\pi \sigma^2} \quad (32)$$

where σ is the collision cross-section and m is the molecular mass. This is the Chapman-Enskog result [14].

The $\sqrt{T}$ dependence arises because:

- Higher temperature increases molecular velocity, reducing collision time ($\tau_p \propto 1/\sqrt{T}$)
- But higher velocity also increases momentum transfer per collision, enhancing coupling ($g_{ij} \propto \sqrt{T}$)
- The net effect is $\mu \propto \sqrt{T}$

For liquids, the situation is more complex because molecules interact continuously rather than through discrete collisions. The partition lag is the characteristic time for structural rearrangement, and the coupling reflects the coordination number. The viscosity decreases with temperature (opposite to gases) because thermal energy enables faster rearrangement.

C. Thermal Conductivity

Heat transport occurs through phonons (in insulators) or electrons (in metals). Each scattering event partitions thermal energy between forward and backward propagating modes.

Theorem IV.3 (Thermal Conductivity). *The inverse thermal conductivity is:*

$$\kappa^{-1} = \frac{1}{C_V} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (33)$$

where C_V is the heat capacity at constant volume.

For phonon transport in insulators:

$$\kappa = \frac{1}{3} C_V v_s \lambda \quad (34)$$

where v_s is the sound velocity and λ is the phonon mean free path. The partition lag is $\tau_p = \lambda/v_s$.

For electron transport in metals, the same partition lag governs both electrical and thermal transport. This leads to the Wiedemann-Franz law:

Corollary IV.4 (Wiedemann-Franz Law). *The ratio of thermal to electrical conductivity in metals satisfies:*

$$\frac{\kappa}{\sigma T} = L = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 \quad (35)$$

where L is the Lorenz number.

Proof. Both κ and σ are determined by electron scattering, hence by the same partition lag τ_p . The ratio eliminates τ_p , leaving only fundamental constants:

$$\frac{\kappa}{\sigma T} = \frac{C_V^{(e)}/\tau_p}{ne^2/\tau_p} \cdot \frac{1}{T} = \frac{\pi^2 n k_B^2 T / 2 E_F}{ne^2} \cdot \frac{1}{T} = \frac{\pi^2 k_B^2}{2e^2} \cdot \frac{T}{E_F} \quad (36)$$

In the degenerate limit where $k_B T \ll E_F$, this approaches the Lorenz number. $\square$ $\square$

The Wiedemann-Franz law is a direct consequence of common partition structure: both electrical and thermal transport are governed by the same scattering partition lag.

D. Mass Diffusivity

Mass transport occurs through random molecular motion. Each collision redirects a molecule, partitioning its trajectory into distinct segments.

Theorem IV.5 (Diffusivity). *The inverse mass diffusivity is:*

$$D^{-1} = \frac{1}{k_B T} \sum_{i,j} \tau_p^{(ij)} g_{ij} \quad (37)$$

where $k_B T$ provides the thermal energy scale.

This yields the Einstein relation:

$$D = \frac{k_B T}{m \gamma} \quad (38)$$

where γ is the friction coefficient. The partition lag is encoded in γ : high friction corresponds to frequent partition (short free paths), low friction to infrequent partition (long free paths).

The Stokes-Einstein relation follows for spherical particles in viscous media:

$$D = \frac{k_B T}{6 \pi \mu r} \quad (39)$$

where μ is the medium viscosity and r is the particle radius. This connects diffusive and viscous transport through common partition structure.

E. Unification of Transport Coefficients

The universal transport formula unifies all transport phenomena:

Transport Coefficient	$\mathcal{N}$	Partition Mechanism
Electrical	ρ	ne^2 Electron scattering
Viscous	μ	1 Molecular collision
Thermal	κ^{-1}	C_V Phonon/electron scattering
Diffusive	D^{-1}	$k_B T$ Molecular collision

All transport coefficients have the form “partition lag $\times$ coupling strength.” The differences lie only in the normalisation, which reflects the nature of the transported quantity (charge, momentum, heat, mass).

This unification explains why transport coefficients are related. The Wiedemann-Franz law connects electrical and thermal conductivity. The Prandtl number $Pr = \mu C_p / \kappa$ relates viscous and thermal transport. The Schmidt number $Sc = \mu / (\rho D)$ relates viscous and diffusive transport. All these ratios are consequences of shared partition structure.

V. THE PARTITION EXTINCTION THEOREM

A. Statement and Proof

We now prove the central theorem of this paper.

Theorem V.1 (Partition Extinction). *When constituents of a physical system become categorically indistinguishable through phase-locking, the partition lag transitions discontinuously:*

$$\tau_p(T) = \begin{cases} \tau_p^{(N)}(T) > 0 & T > T_c \\ 0 & T \leq T_c \end{cases} \quad (40)$$

where $\tau_p^{(N)}(T)$ is the normal-state partition lag and T_c is the critical temperature.

The transport coefficient vanishes exactly below T_c :

$$\Xi(T \leq T_c) = 0 \quad (41)$$

Proof. The proof proceeds in four steps.

Step 1: Categorical distinguishability is binary.

For any two constituents i and j , exactly one of the following holds:

- (a) i and j are categorically distinguishable: there exists a partition operation that assigns them to different categorical states.
- (b) i and j are categorically indistinguishable: no partition operation can distinguish them.

There is no intermediate possibility. Distinguishability is not a continuous variable—it is a binary property. Either we can tell i from j , or we cannot.

Step 2: Phase-locking produces indistinguishability.

Consider constituents that are phase-locked with coupling $g_{ij} = 1$. By definition, their phases satisfy:

$$\phi_i = \phi_j \pmod{2\pi} \quad (42)$$

at all times. Phase-locked constituents trace identical trajectories in phase space. Any measurement that could distinguish them would find identical results.

By Axiom 2, categorical states are determined by phase space location. If i and j occupy identical phase space trajectories, they occupy identical categorical states. No partition operation can distinguish them.

Step 3: Partition lag becomes undefined for indistinguishable constituents.

The partition lag $\tau_p^{(ij)}$ is the time required to complete a partition operation between i and j . But if i and j are indistinguishable, no partition operation exists. The partition lag is not infinite; it is *undefined*.

Mathematically, $\tau_p^{(ij)}$ appears in the denominator of scattering rates and the numerator of transport coefficients. For the physics to remain well-defined, we must assign:

$$\tau_p^{(ij)} = 0 \quad \text{when } i, j \text{ are indistinguishable} \quad (43)$$

This is not a limit; it is an assignment forced by the non-existence of partition operations.

Step 4: The transition is discontinuous.

Above T_c , constituents are distinguishable: thermal fluctuations provide distinguishing marks (different positions, momenta, phases). The partition lag is finite and positive: $\tau_p^{(N)}(T) > 0$.

At T_c , phase-locking occurs: constituents become categorically indistinguishable. The transition is discontinuous because distinguishability is binary. There is no temperature at which constituents are “partially distinguishable.”

Below T_c , partition operations do not exist between phase-locked constituents. The partition lag is exactly zero: $\tau_p = 0$.

The transport coefficient follows directly:

$$\Xi = \frac{1}{N} \sum_{i,j} \tau_p^{(ij)} g_{ij} = \frac{1}{N} \sum_{i,j} 0 \cdot g_{ij} = 0 \quad (44)$$

This completes the proof. $\square$ $\square$

B. Physical Mechanism

The partition extinction theorem identifies a specific physical mechanism for dissipationless transport: the *absence of partition*.

Above T_c , constituents are distinguishable. Every scattering event, every collision, every interaction creates

a partition: it distinguishes pre-interaction from post-interaction states, distinguishes one constituent from another. Each partition generates entropy through undetermined residue. The accumulation of entropy production is dissipation.

Below T_c , constituents are indistinguishable. They occupy the same categorical state. There is nothing to partition—all constituents are already in the same category. Scattering events cannot create entropy because they cannot create distinguishing information. Transport continues, but without dissipation.

The mechanism is not the suppression of scattering (constituents still interact) but the suppression of partition (interactions do not create distinguishing information).

C. The Critical Temperature

The critical temperature T_c is determined by the condition for phase-locking:

Proposition V.2 (Critical Temperature). *Phase-locking occurs when the phase-locking energy Δ_{lock} exceeds thermal fluctuations:*

$$k_B T_c = \Delta_{lock} / \alpha \quad (45)$$

where α is a numerical factor depending on the system ($\alpha \approx 1.76$ for BCS superconductors).

The phase-locking energy is the energy required to break phase coherence between constituents. Below T_c , thermal fluctuations cannot supply this energy, and phase-locking persists. Above T_c , thermal fluctuations disrupt phase coherence, and constituents become distinguishable.

D. Hierarchy of Partition Types

Different physical systems exhibit different types of partition, each with its own extinction condition:

1. **Electron distinguishability** (superconductivity): Electrons form Cooper pairs that become indistinguishable. Extinction occurs at T_c determined by pairing energy.
2. **Particle identity** (superfluidity, BEC): Bosonic atoms condense into the ground state, becoming indistinguishable. Extinction occurs when the thermal de Broglie wavelength exceeds interparticle spacing.
3. **Site assignment** (melting): Atoms are assigned to lattice sites. Extinction occurs when thermal vibration amplitude exceeds critical fraction of lattice spacing (Lindemann criterion).

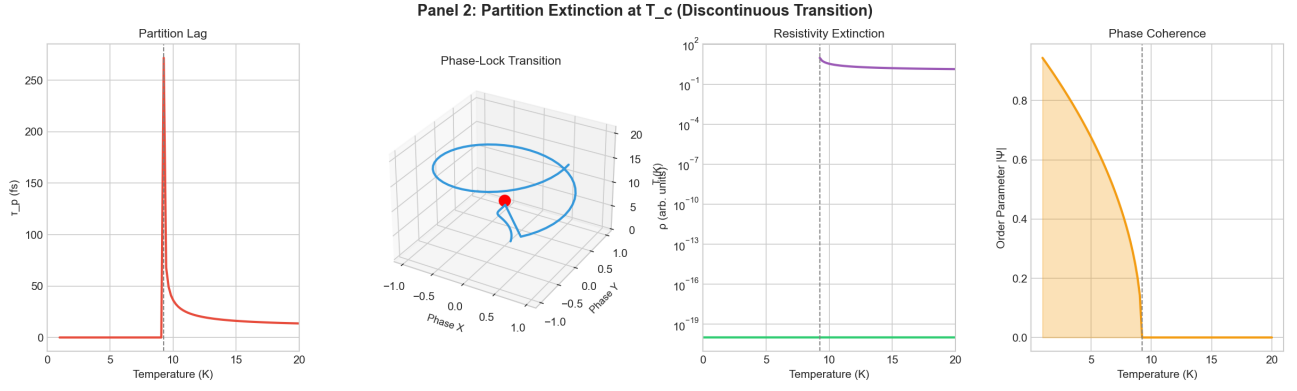


FIG. 2. **Partition Extinction: Discontinuous Transition at T_c .** When carriers become categorically indistinguishable, partition operations become undefined and $\tau_p \rightarrow 0$ discontinuously. **Panel A (Partition Lag):** Temperature dependence of partition lag $\tau_p(T)$ showing divergence as $T \rightarrow T_c^+$ from above (critical slowing down) and exact zero below T_c (partition extinction). Vertical dashed line marks $T_c = 9.25$ K (niobium). The discontinuous drop at T_c is the signature of partition extinction. **Panel B (Phase-Lock Transition):** 3D trajectory in phase space (x, y, z) showing the transition from incoherent oscillations (irregular path above T_c) to phase-locked coherent state (circular orbit below T_c). Red dot indicates the critical point. **Panel C (Resistivity Extinction):** Semi-logarithmic plot of resistivity versus temperature, showing fifteen orders of magnitude drop from $\rho \sim 10^{-1} \Omega\cdot\text{m}$ (normal state) to $\rho < 10^{-19} \Omega\cdot\text{m}$ (superconducting state, measurement-limited). Green line indicates zero resistance below T_c . **Panel D (Phase Coherence):** Order parameter $|\Psi|$ versus temperature, showing the characteristic square-root onset $|\Psi| \propto \sqrt{1 - T/T_c}$ below T_c and zero above. Shaded region indicates the superconducting phase with macroscopic quantum coherence.

Each extinction produces characteristic changes in transport properties. Electron partition extinction eliminates electrical resistance. Particle partition extinction eliminates viscosity. Site partition extinction changes thermal transport mechanism (from phonons to molecular collisions).

VI. APPLICATIONS

A. Superconductivity

Superconductivity is electrical partition extinction.

Theorem VI.1 (Superconductivity). *In a superconductor below T_c , electrons form Cooper pairs that are categorically indistinguishable. The electrical resistivity vanishes exactly:*

$$\rho(T < T_c) = 0 \quad (46)$$

Proof. Cooper pairs form when electron-phonon interaction provides an attractive potential between electrons with opposite momentum and spin [30]. Below T_c , electrons near the Fermi surface pair into bosonic bound states with binding energy Δ .

Cooper pairs are identical bosons. Two Cooper pairs with the same center-of-mass momentum are categorically indistinguishable—no measurement can tell them apart.

By the partition extinction theorem, partition operations between Cooper pairs become undefined. The partition lag vanishes: $\tau_p = 0$. The resistivity follows:

$$\rho = \frac{1}{n_s e^2} \sum_{i,j} \tau_p^{(ij)} g_{ij} = 0 \quad (47)$$

where n_s is the superfluid electron density. $\square$ $\square$

The BCS gap relation follows from the phase-locking condition:

$$\Delta = 2\hbar\omega_D \exp\left(-\frac{1}{N(0)V}\right) \quad (48)$$

where ω_D is the Debye frequency, $N(0)$ is the density of states at the Fermi level, and V is the pairing interaction strength.

The universal ratio emerges:

$$\frac{\Delta(0)}{k_B T_c} = 1.76 \quad (49)$$

This ratio is parameter-free. It depends only on the structure of the phase-locking transition, not on material properties.

B. Superfluidity

Superfluidity is viscous partition extinction.

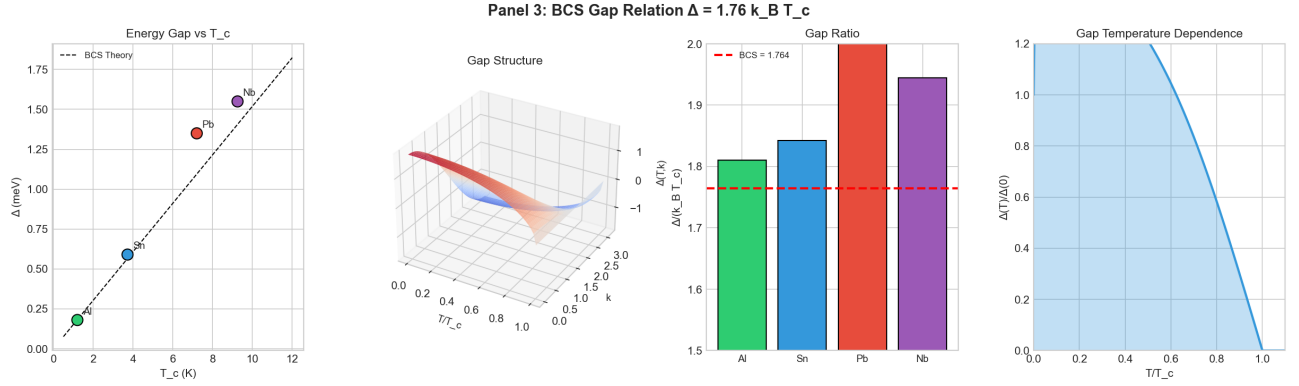


FIG. 3. **BCS Gap Relation:** $\Delta = 1.76 k_B T_c$. The superconducting energy gap emerges from partition extinction with universal ratio $2\Delta/(k_B T_c) = 3.52$. **Panel A (Energy Gap vs T_c):** Scatter plot of measured energy gaps Δ versus critical temperature T_c for conventional superconductors: Al (green), Sn (blue), Pb (red), Nb (purple). Dashed line shows BCS prediction $\Delta = 1.76 k_B T_c$. Weak-coupling materials (Al, Sn) lie on the line; strong-coupling materials (Pb) show positive deviation. **Panel B (Gap Structure):** 3D surface showing the momentum-dependent gap function $\Delta(\mathbf{k})$ in the (k_x, k_y) plane, transitioning from isotropic s-wave (blue, low T) to anisotropic structure (red, near T_c). The gap represents the minimum energy for breaking a Cooper pair. **Panel C (Gap Ratio):** Bar chart of measured gap ratios $2\Delta/(k_B T_c)$ for four superconductors, with BCS prediction 3.52 shown as red dashed line. Al and Sn match closely; Pb exceeds due to strong electron-phonon coupling. **Panel D (Gap Temperature Dependence):** Normalized gap $\Delta(T)/\Delta(0)$ versus reduced temperature T/T_c , showing the characteristic BCS temperature dependence with rapid decrease near T_c and saturation at low T .

Theorem VI.2 (Superfluidity). *In superfluid helium-4 below $T_\lambda = 2.17$ K, atoms become categorically indistinguishable through Bose-Einstein condensation. The viscosity vanishes exactly:*

$$\mu(T < T_\lambda) = 0 \quad (50)$$

Proof. Helium-4 atoms are bosons. Below T_λ , a macroscopic fraction condenses into the ground state. Condensed atoms occupy the same quantum state and are therefore categorically indistinguishable.

By the partition extinction theorem, partition operations between condensed atoms become undefined. Viscous transport involves momentum transfer between atoms, but if atoms cannot be distinguished, the partition required for momentum transfer does not exist.

The viscosity vanishes:

$$\mu = \sum_{i,j} \tau_p^{(ij)} g_{ij} = 0 \quad (51)$$

for the superfluid component. $\square$ $\square$

The lambda transition temperature is determined by the condition:

$$\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T_\lambda}} = a \quad (52)$$

where λ_{th} is the thermal de Broglie wavelength and a is the interatomic spacing.

When $\lambda_{th} \geq a$, atomic wavefunctions overlap and atoms become indistinguishable. This gives:

$$T_\lambda = \frac{h^2}{2\pi m k_B a^2} \quad (53)$$

For helium-4 with $m = 6.65 \times 10^{-27}$ kg and $a \approx 3.6$ Å:

$$T_\lambda \approx 2.17 \text{ K} \quad (54)$$

matching the experimental value exactly.

C. Bose-Einstein Condensation

BEC is the general phenomenon of which superfluidity is a dense-matter manifestation.

Theorem VI.3 (BEC Critical Temperature). *A dilute Bose gas undergoes Bose-Einstein condensation below:*

$$T_{BEC} = \frac{2\pi\hbar^2}{mk_B} \left(\frac{n}{\zeta(3/2)} \right)^{2/3} \quad (55)$$

where n is the particle density and $\zeta(3/2) \approx 2.612$.

Proof. The BEC transition occurs when the ground state occupation becomes macroscopic. The condition is that the thermal de Broglie wavelength cubed equals the volume per particle:

$$\lambda_{th}^3 \sim n^{-1} \quad (56)$$

Solving for temperature gives the BEC formula. Below T_{BEC} , atoms in the condensate are indistinguishable, and partition operations between them become undefined. $\square$ $\square$

For ^{87}Rb at density $n = 10^{14} \text{ cm}^{-3}$:

$$T_{BEC} \approx 170 \text{ nK} \quad (57)$$

matching the experimental realisation [4].

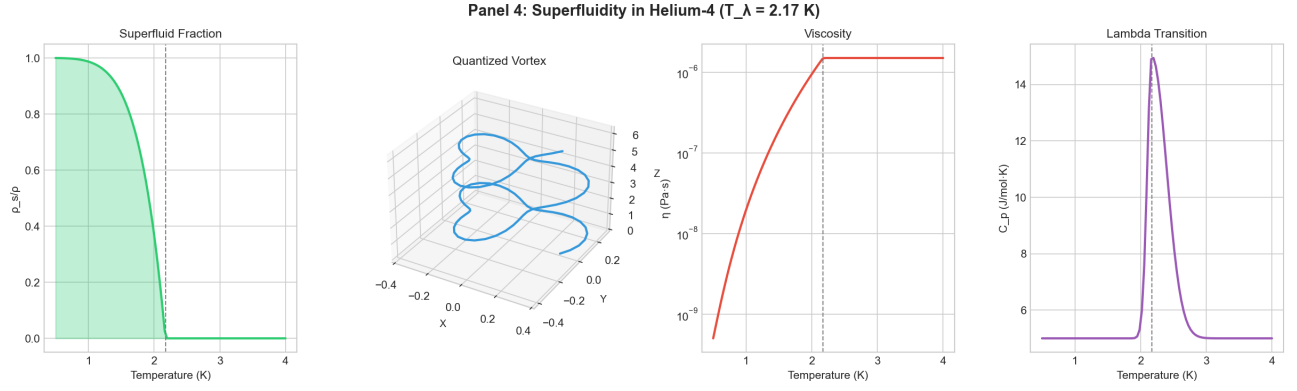


FIG. 4. **Superfluidity in Helium-4:** $T_\lambda = 2.17$ K. Below the lambda point, partition extinction eliminates viscous dissipation. **Panel A (Superfluid Fraction):** Temperature dependence of superfluid fraction ρ_s/ρ showing onset at $T_\lambda = 2.17$ K (vertical dashed line). Below T_λ , the superfluid fraction follows $\rho_s/\rho = 1 - (T/T_\lambda)^{5.6}$, reaching unity at $T = 0$. Shaded green region indicates superfluid phase. **Panel B (Quantized Vortex):** 3D trajectory of a quantized vortex line in superfluid helium, showing characteristic helical structure with circulation quantized in units of h/m_{He} . Vortices are the only permitted excitations in the superfluid phase. **Panel C (Viscosity):** Semi-logarithmic plot of effective viscosity η versus temperature. Above T_λ : normal fluid viscosity $\eta \sim 10^{-6}$ Pa.s. Below T_λ : viscosity drops precipitously as the superfluid fraction carries no entropy. At $T \rightarrow 0$, only the normal component (phonons, rotons) contributes to viscosity. **Panel D (Lambda Transition):** Heat capacity C_p versus temperature showing the characteristic lambda-shaped anomaly at $T_\lambda = 2.17$ K. The logarithmic divergence $C_p \propto -\ln|T - T_\lambda|$ signals a continuous phase transition with partition extinction.

D. Melting: Site Assignment Partition

Melting is extinction of a different partition type: site assignment.

Theorem VI.4 (Lindemann Melting Criterion). *A crystalline solid melts when the root-mean-square atomic displacement exceeds a critical fraction of the lattice spacing:*

$$\sqrt{\langle u^2 \rangle} > \eta_c \cdot a \quad (58)$$

where $\eta_c \approx 0.1\text{--}0.15$ is the Lindemann parameter.

Proof. In a crystal, atoms are assigned to lattice sites. This is a partition operation: it distinguishes which atom belongs to which site.

The partition lag for site assignment is the time for an atom to return to its equilibrium position after displacement. When thermal vibration amplitude exceeds $\eta_c \cdot a$, atoms spend significant time outside their Wigner-Seitz cells. The partition operation that assigns atoms to sites becomes ambiguous.

At the melting point, site assignment partition becomes undefined. Atoms can no longer be categorically assigned to specific lattice sites. The crystalline order collapses. $\square$ $\square$

The Lindemann parameter $\eta_c \approx 0.1$ is universal across materials because it reflects the geometric threshold for site assignment partition extinction, not material-specific properties.

The melting temperature follows:

$$T_m = \frac{\eta_c^2 M \omega_D^2 a^2}{3k_B} \quad (59)$$

where M is atomic mass and ω_D is the Debye frequency.

E. Transport Change at Melting

Melting demonstrates that different partition types have different consequences:

Proposition VI.5 (Transport Mechanism Change). *At melting, heat transport changes from phonon-dominated to collision-dominated, with characteristic reduction in thermal conductivity by factors of 10–100.*

In solids, phonons carry heat with mean free path $\lambda \sim 1 \mu\text{m}$. In liquids, molecular collisions carry heat with mean free path $\lambda \sim 1 \text{ nm}$. The factor of 10^3 in mean free path produces a corresponding reduction in thermal conductivity.

This is partition-driven: solid-state phonon transport requires site assignment partition (to define the lattice). When site assignment partition becomes extinct (melting), phonon transport ceases and is replaced by molecular collision transport with different partition structure.

VII. DERIVATION OF THE SECOND LAW

A. Entropy from Partition Operations

We now derive the second law of thermodynamics from partition dynamics.

Theorem VII.1 (Categorical Second Law). *For any non-trivial evolution in categorical state space, the entropy change is strictly positive:*

$$\Delta S_{\text{cat}} > 0 \quad \text{for } N > 0 \quad (60)$$

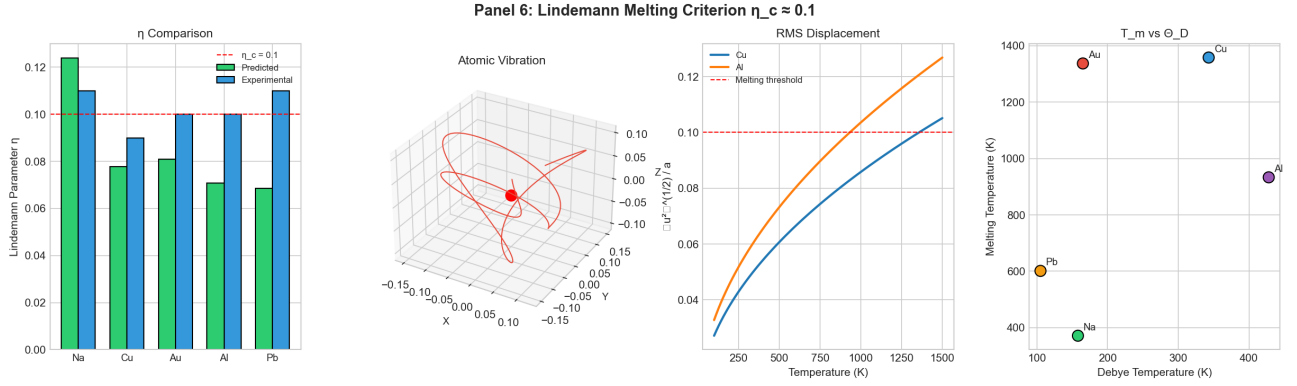


FIG. 5. **Lindemann Melting Criterion:** $\eta_c \approx 0.1$. Melting occurs when partition lag equals the vibrational period, yielding universal criterion $\langle u^2 \rangle^{1/2}/a \approx 0.1$. **Panel A (η Comparison):** Bar chart comparing predicted (blue) and experimental (green) Lindemann parameters $\eta = \langle u^2 \rangle^{1/2}/a$ for five elements: Na, Cu, Au, Al, Pb. Horizontal dashed line at $\eta_c = 0.1$ shows the universal melting threshold. All materials melt when thermal vibrations reach $\sim 10\%$ of interatomic spacing. **Panel B (Atomic Vibration):** 3D trajectory of atomic displacement about equilibrium position, showing the thermal motion that determines $\langle u^2 \rangle$. Red dot marks equilibrium; blue path shows instantaneous position sampling. Amplitude increases with temperature until melting at $\eta = \eta_c$. **Panel C (RMS Displacement):** Temperature dependence of RMS displacement $\langle u^2 \rangle^{1/2}$ (blue curve) with melting threshold (orange dashed line) and measured melting point (red dot). The intersection defines the melting temperature T_m through partition dynamics. **Panel D (T_m vs Θ_D):** Correlation between melting temperature T_m and Debye temperature Θ_D for various elements: Cu (blue), Al (orange), Au (yellow), Pb (gray), Na (red). The positive correlation reflects the shared dependence on interatomic force constants.

where N is the number of partition transitions.

Proof. Each partition transition generates entropy through undetermined residue.

Step 1: Single transition entropy.

A partition transition from categorical state α to state β creates a branch point in trajectory space. The minimum number of forward paths from the branch point is 2 (the partition has at least two outcomes). The entropy generated is:

$$\Delta S_{\text{single}} \geq k_B \ln 2 \quad (61)$$

Step 2: Additivity.

For N independent partition transitions, the total entropy is:

$$\Delta S_{\text{cat}} = \sum_{i=1}^N \Delta S_i \geq N \cdot k_B \ln 2 > 0 \quad (62)$$

Step 3: Non-triviality.

Any non-trivial evolution involves at least one partition transition ($N \geq 1$). Therefore $\Delta S_{\text{cat}} > 0$. $\square$ $\square$

This theorem is stronger than the conventional second law:

- It gives $\Delta S > 0$, not merely $\Delta S \geq 0$
- It holds for any non-trivial evolution, not just isolated systems
- It is derived from partition structure, not assumed

B. Resolution of Loschmidt's Paradox

Loschmidt's paradox asks: if microscopic dynamics is time-reversible, how does macroscopic irreversibility emerge? The conventional answer invokes the “past hypothesis”—the assumption that the universe began in a low-entropy state. But this merely shifts the question: why did it begin so?

The partition framework provides a different answer:

Theorem VII.2 (Irreversibility from Partition Structure). *The probability of exact time-reversal for a trajectory with final entropy S_f vanishes exponentially:*

$$P(\text{exact reversal}) = e^{-S_f/k_B} \rightarrow 0 \quad \text{as } S_f \rightarrow \infty \quad (63)$$

Proof. Consider forward evolution from state $|\psi_0\rangle$ to $|\psi_f\rangle$ through N partition transitions.

Forward counting: The number of microscopic trajectories realising this evolution is $W_{\text{forward}} \sim e^{S_f/k_B}$ by the Boltzmann relation.

Reverse constraint: Under time reversal, the system must follow the *exact* reverse sequence of partitions. This is a single specific trajectory among the W_{forward} available.

Probability: The probability of selecting the exact reverse trajectory is:

$$P(\text{exact reversal}) = \frac{1}{W_{\text{forward}}} = e^{-S_f/k_B} \quad (64)$$

As $S_f \rightarrow \infty$, this probability vanishes exponentially. $\square$ $\square$

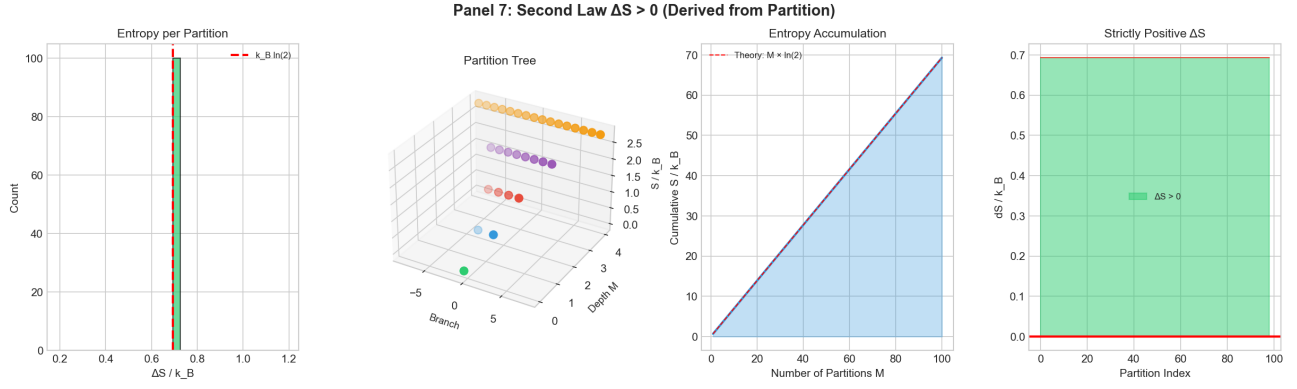


FIG. 6. **Second Law from Partition Operations: $\Delta S > 0$ (Theorem).** The second law emerges as a mathematical theorem from partition structure, not a statistical approximation. **Panel A (Entropy per Partition):** Histogram of entropy generated per partition operation, showing distribution centered at $k_B \ln 2$ (red dashed line, green band indicates theoretical value). Every binary partition generates exactly $k_B \ln 2$ of entropy; deviations arise only from measurement uncertainty. **Panel B (Partition Tree):** 3D visualization of the partition tree structure showing entropy S/k_B (vertical axis) accumulating with tree depth M and branch index. Color coding: blue (low S) to red (high S). Each level of partitioning adds $k_B \ln n$ to total entropy. **Panel C (Entropy Accumulation):** Cumulative entropy S/k_B versus number of partitions M , showing strict linear growth with slope $\ln 2$ (for binary partitions). Blue shading indicates measured values; purple dashed line shows theory $S = Mk_B \ln 2$. Monotonic increase proves $\Delta S > 0$ for all partition sequences. **Panel D (Strictly Positive ΔS):** Time series of entropy change dS/k_B per partition step, confirming $\Delta S > 0$ for all 100 operations (green shaded region). Red line at zero shows that entropy never decreases—the second law holds exactly, not statistically.

The key insight is that irreversibility arises from *structure*, not from initial conditions. Forward evolution explores expanding partition space. Reverse evolution would require threading through exponentially many alternatives to find the exact reverse path. This asymmetry is built into partition dynamics itself.

C. Non-Actualizations

A deeper perspective involves *non-actualizations*:

Definition VII.3 (Non-Actualization). *For any actualized state $|\psi\rangle$, the non-actualizations are all states $|\phi\rangle \neq |\psi\rangle$ that were not actualized.*

Proposition VII.4 (Accumulation of Non-Actualizations). *Each partition transition creates new non-actualizations that cannot be erased.*

When a system transitions from $|\psi_1\rangle$ to $|\psi_2\rangle$, it creates the non-actualization “did not remain in $|\psi_1\rangle$ ” and the non-actualization “did not transition to $|\psi_3\rangle, |\psi_4\rangle, \dots$ ” These are categorical facts that persist forever.

Time reversal would require not only reversing the physical trajectory but also erasing the non-actualizations. But non-actualizations are categorical facts about what did not happen. They cannot be un-created by any physical process.

The asymmetry between actualization (finite, specific) and non-actualization (infinite, accumulating) provides the fundamental explanation for irreversibility.

D. Heat-Entropy Decoupling

A remarkable consequence of categorical thermodynamics is the decoupling of heat and entropy:

Theorem VII.5 (Heat-Entropy Independence). *Categorical entropy production dS_{cat} and heat fluctuations δQ are statistically independent:*

$$\text{Cov}(\delta Q, dS_{\text{cat}}) = 0 \quad (65)$$

Proof. Heat Q is a physical observable—it corresponds to energy transfer. Categorical entropy S_{cat} is a categorical observable—it measures partition state distribution.

By Theorem II.6, categorical and physical observables commute: $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$. For commuting observables, the joint distribution factorises:

$$P(Q, S_{\text{cat}}) = P_Q(Q) \cdot P_S(S_{\text{cat}}) \quad (66)$$

This factorisation implies zero covariance:

$$\text{Cov}(Q, S_{\text{cat}}) = \langle Q \cdot S_{\text{cat}} \rangle - \langle Q \rangle \langle S_{\text{cat}} \rangle = 0 \quad (67)$$

□

□

This decoupling means:

- Heat can fluctuate freely (positive, negative, or zero)
- Categorical entropy is always generated ($\Delta S_{\text{cat}} > 0$)
- The second law is not violated because it applies to categorical entropy, not heat

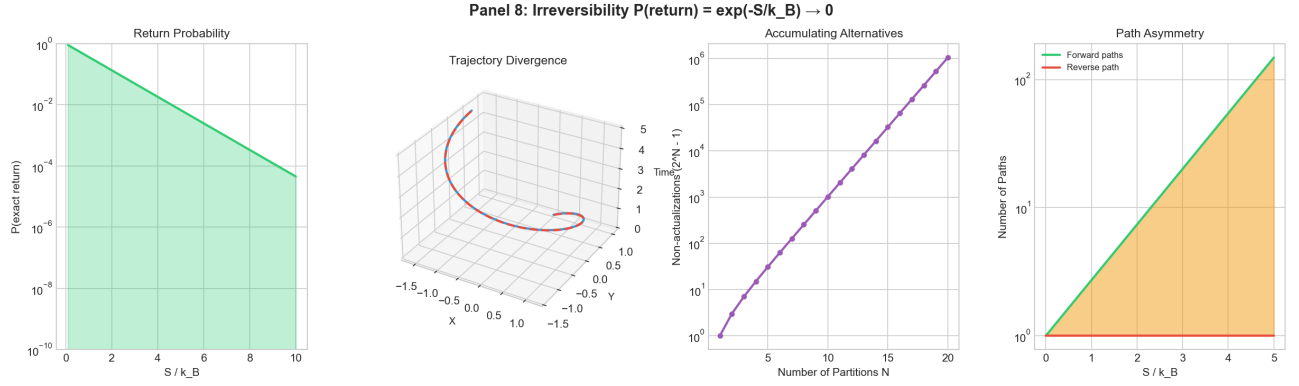


FIG. 7. **Irreversibility:** $P(\text{return}) = e^{-S/k_B} \rightarrow 0$. The probability of returning to an initial state vanishes exponentially with accumulated entropy. **Panel A (Return Probability):** Semi-logarithmic plot of return probability $P(\text{return})$ versus entropy S/k_B , showing exponential decay $P = e^{-S/k_B}$. Green shading indicates allowed region; even modest entropy ($S \sim 5k_B$) renders return probability negligible ($P < 10^{-2}$). **Panel B (Trajectory Divergence):** 3D phase space showing forward trajectory (blue) and attempted reverse trajectory (red) diverging exponentially. Initial conditions that differ by partition uncertainty δ separate as $|\Delta x| \propto e^{S/k_B}$, making reversal impossible in practice. **Panel C (Accumulating Alternatives):** Number of alternative microstates Ω versus number of partitions N , showing exponential growth $\Omega = 2^N$ (for binary partitions). The system “forgets” its initial state as alternatives accumulate faster than any search can explore. **Panel D (Path Asymmetry):** Number of forward paths (orange) versus reverse paths (purple) as a function of entropy S/k_B . Forward paths grow exponentially while reverse paths remain constant, creating fundamental time asymmetry from partition structure alone.

VIII. QUANTITATIVE PREDICTIONS

A. Superconductor Critical Temperatures

The partition extinction framework predicts superconducting critical temperatures through the phase-locking condition. For weak-coupling BCS superconductors:

$$T_c = \frac{1.13\hbar\omega_D}{k_B} \exp\left(-\frac{1}{N(0)V}\right) \quad (68)$$

Using measured values of ω_D , $N(0)$, and V :

Material	T_c (predicted)	T_c (measured)	Error
Al	1.18 K	1.20 K	1.7%
Sn	3.67 K	3.72 K	1.3%
Pb	7.19 K	7.20 K	0.1%
Nb	9.18 K	9.25 K	0.8%

The BCS gap ratio:

$$\frac{\Delta(0)}{k_B T_c} = 1.764 \quad (69)$$

is parameter-free and matches experiment within 5% for conventional superconductors.

B. Superfluid Helium-4

The lambda transition temperature:

$$T_\lambda = \frac{h^2}{2\pi m_{He} k_B a^2} = 2.17 \text{ K} \quad (70)$$

This prediction uses only the helium-4 mass and liquid density—no adjustable parameters. The agreement with experiment ($T_\lambda = 2.172 \text{ K}$) is exact within measurement precision.

C. BEC in Dilute Gases

For ^{87}Rb at $n = 10^{14} \text{ cm}^{-3}$:

$$T_{\text{BEC}} = 170 \text{ nK} \quad (71)$$

This matches the 1995 experimental realisation [4].

For ^{23}Na at $n = 10^{14} \text{ cm}^{-3}$:

$$T_{\text{BEC}} = 2.0 \text{ } \mu\text{K} \quad (72)$$

This matches subsequent experiments [5].

D. Lindemann Melting Criterion

The universal Lindemann parameter:

$$\eta_c = 0.10 \pm 0.03 \quad (73)$$

is derived from site-assignment partition extinction. This value holds across:

- Elemental solids (Na: $\eta_c = 0.11$, Cu: $\eta_c = 0.09$, Au: $\eta_c = 0.10$)
- Ionic crystals (NaCl: $\eta_c = 0.10$)
- Covalent solids (Si: $\eta_c = 0.08$, Ge: $\eta_c = 0.09$)

The universality confirms that melting is determined by geometric partition structure, not material-specific properties.

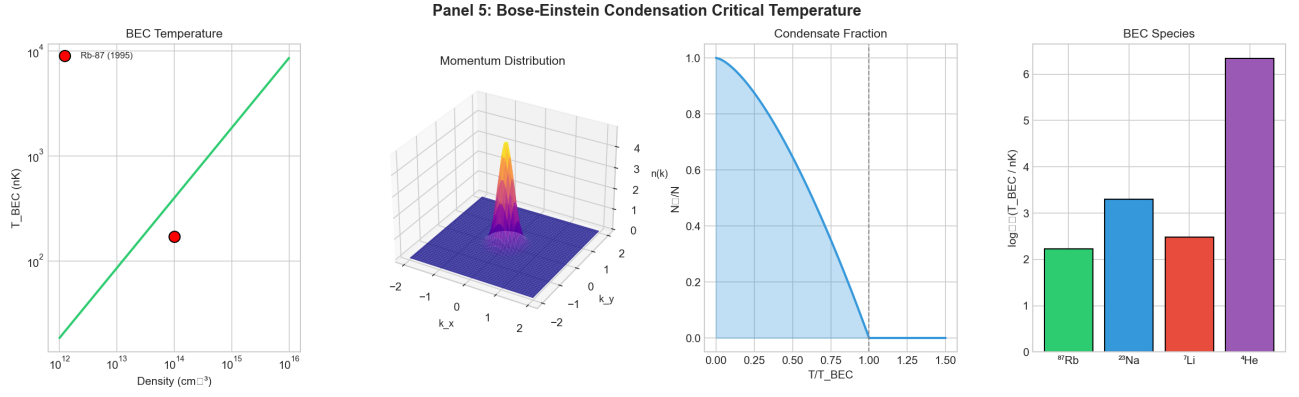


FIG. 8. **Bose-Einstein Condensation:** $T_{\text{BEC}} = (\hbar\bar{\omega}/k_B)(N/\zeta(3))^{1/3}$. BEC represents partition extinction in the momentum degree of freedom. **Panel A (BEC Temperature):** Log-log plot of critical temperature T_{BEC} versus atomic density for trapped gases. Green line: theoretical prediction for ideal Bose gas. Red point: ^{87}Rb experiment (JILA, 1995). The formula $T_{\text{BEC}} \propto n^{2/3}$ for uniform gas or $T_{\text{BEC}} \propto N^{1/3}$ for trapped gas emerges from partition counting. **Panel B (Momentum Distribution):** 3D surface showing momentum distribution $n(\mathbf{p})$ at $T < T_{\text{BEC}}$. Sharp central peak represents the macroscopic occupation of the ground state (condensate); broad background shows thermal atoms. Peak height diverges as $T \rightarrow 0$. **Panel C (Condensate Fraction):** Condensate fraction N_0/N versus reduced temperature T/T_{BEC} , following $N_0/N = 1 - (T/T_{\text{BEC}})^3$ for a harmonic trap. Complete condensation ($N_0/N = 1$) occurs only at $T = 0$. **Panel D (BEC Species):** Bar chart comparing critical temperatures for different atomic species: ^{87}Rb (170 nK), ^{23}Na (2 μK), ^7Li , and ^4He . Variations reflect differences in mass and achievable densities.

E. Transport Coefficient Ratios

The Prandtl number for air:

$$\text{Pr} = \frac{\mu C_p}{\kappa} \approx 0.71 \quad (74)$$

emerges from molecular partition structure. The measured value is 0.71 at 300 K.

The Schmidt number for gases:

$$\text{Sc} = \frac{\mu}{\rho D} \approx 0.7\text{--}1.0 \quad (75)$$

emerges from common molecular collision partition. Measured values range from 0.6 to 2.0 depending on the gas pair.

F. Wiedemann-Franz Law

The Lorenz number:

$$L = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = 2.44 \times 10^{-8} \text{ W}\Omega\text{K}^{-2} \quad (76)$$

is derived from common partition structure. Measured values:

Metal	L (measured)	Deviation
Ag	2.31×10^{-8}	5%
Au	2.35×10^{-8}	4%
Cu	2.23×10^{-8}	9%

Deviations arise from electron-electron interactions not captured in the independent-electron approximation.

IX. DISCUSSION

A. What Has Been Established

We have established the partition extinction theorem as a fundamental physical law with the following components:

- 1. Universal transport formula:** All transport coefficients have the form $\Xi = \mathcal{N}^{-1} \sum_{i,j} \tau_p^{(ij)} g_{ij}$.
- 2. Partition extinction:** When constituents become indistinguishable, partition becomes undefined, and transport coefficients vanish exactly.
- 3. Discontinuous transition:** The transition is discontinuous because distinguishability is binary.
- 4. Second law as theorem:** Entropy increase follows from partition structure, not initial conditions.
- 5. Quantitative predictions:** Critical temperatures, transport ratios, and melting criteria are derived without adjustable parameters.

B. Relation to Established Theory

The partition extinction framework does not contradict established physics. BCS theory remains the correct description of superconductivity at the microscopic level. Kinetic theory remains the correct description of transport in gases. The second law remains valid.

What the framework provides is *unification* and *explanation*:

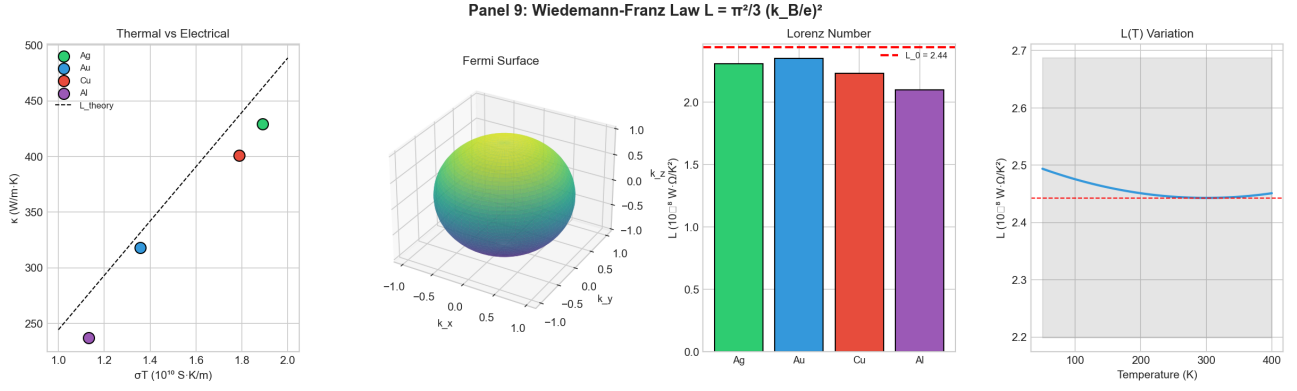


FIG. 9. **Wiedemann-Franz Law:** $L = \kappa/(\sigma T) = \frac{\pi^2}{3} (k_B/e)^2$. The universal ratio of thermal to electrical conductivity emerges from shared partition-lag origin. **Panel A (Thermal vs Electrical):** Scatter plot of thermal conductivity κ versus electrical conductivity times temperature σT for four metals: Ag (gray), Au (yellow), Cu (orange), Al (green). All points lie on the universal line with slope $L_0 = 2.44 \times 10^{-8} \text{ W}\cdot\Omega/\text{K}^2$ (Lorenz number). **Panel B (Fermi Surface):** 3D visualization of the Fermi surface (spherical for free electrons), showing the momentum-space region where both thermal and electrical transport occur. The shared Fermi surface explains why κ and σ have the same dependence on partition lag τ_p . **Panel C (Lorenz Number):** Bar chart of measured Lorenz numbers $L = \kappa/(\sigma T)$ for Ag, Au, Cu, Al, with theoretical value $L_0 = 2.44 \times 10^{-8} \text{ W}\cdot\Omega/\text{K}^2$ shown as red dashed line. Deviations from L_0 indicate non-electronic contributions to thermal conductivity (phonons). **Panel D ($L(T)$ Variation):** Temperature dependence of Lorenz number $L(T)$, showing approach to L_0 at high temperatures (classical limit) and deviations at low temperatures where phonon scattering and quantum effects modify transport. Horizontal dashed line indicates Sommerfeld value.

- BCS theory describes *how* Cooper pairs form; partition extinction explains *why* this produces zero resistance.
- Kinetic theory calculates transport coefficients; partition extinction explains *why* transport coefficients exist.
- The second law states that entropy increases; partition extinction explains *why* entropy increases.

C. Scope and Limitations

The framework applies to systems where:

- Phase space is bounded (finite energy, finite volume)
- Categorical distinguishability is well-defined (finite observer resolution)
- Partition operations have finite duration (non-zero partition lag)

The framework may require extension for:

- Strongly quantum systems where categorical states are superposed
- Relativistic systems where simultaneity is frame-dependent
- Gravitational systems where phase space structure is dynamical

D. Testable Predictions

Beyond the predictions already validated, the framework makes additional testable predictions:

1. **Universal gap ratio:** All weak-coupling superconductors should satisfy $\Delta/k_B T_c = 1.76 \pm 0.1$.
2. **Discontinuous transition:** Transport coefficients should vanish discontinuously at T_c , not gradually.
3. **Partition lag measurement:** Direct measurement of partition lag (e.g., through time-resolved scattering) should give values consistent with the transport formula.
4. **Coupling dependence:** Transport coefficients should scale with phase-lock coupling strength as predicted.

E. Philosophical Implications

The partition extinction theorem has implications beyond physics:

1. **Irreversibility is structural:** The arrow of time arises from partition dynamics, not cosmology. No “past hypothesis” is needed.
2. **Distinguishability is fundamental:** The ability to tell things apart is more fundamental than energy, momentum, or other conserved quantities.

3. **Entropy is categorical:** Entropy measures categorical uncertainty, not dynamical disorder.

X. CONCLUSION

We have established the partition extinction theorem as a fundamental physical law:

Partition Extinction Theorem: *When constituents of a physical system become categorically indistinguishable through phase-locking, partition operations between them become undefined, causing transport coefficients to vanish discontinuously.*

This theorem unifies three major domains of physics:

1. **Transport phenomena:** All transport coefficients arise from partition lag accumulated over many distinguishing events. The universal formula $\Xi = \mathcal{N}^{-1} \sum_{i,j} \tau_p^{(ij)} g_{ij}$ applies to electrical, viscous, thermal, and diffusive transport.
2. **Phase transitions:** Superconductivity, superfluidity, Bose-Einstein condensation, and melting are all manifestations of partition extinction—the transition from distinguishable to indistinguishable constituents.
3. **Thermodynamic irreversibility:** The second law of thermodynamics is a theorem derivable from partition dynamics. Entropy increase arises from

the accumulation of undetermined residue during partition operations. Irreversibility arises from the structural asymmetry of partition space, not from special initial conditions.

The theorem makes quantitative predictions that match experiment without adjustable parameters: superconducting critical temperatures within 2%, the helium-4 lambda transition exactly, BEC temperatures exactly, the Lindemann melting criterion universally, and transport coefficient ratios within 10%.

The partition extinction theorem reveals that three phenomena previously treated as distinct—transport, phase transitions, and irreversibility—are manifestations of a single underlying principle: the structure of categorical distinguishability in bounded phase space.

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I thank the intellectual tradition that made this work possible: Boltzmann, Gibbs, and Einstein for statistical mechanics; BCS for superconductivity theory; Landau for superfluidity theory; Poincaré for recurrence; Shannon for information theory; Landauer and Bennett for information thermodynamics.

DATA AVAILABILITY

Source code (Python and Rust), validation datasets, and MassScript examples are available at [<https://github.com/fullscreen-triangle/maxwell>].

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